QUESTION BANK

Course Details		
Class: IV - Year, B.Tech	**Semester:** I	**Academic Year:** 2025-26
Course Title: Digital Image Processing	**Course Code: 21EC2019**	**Credits:** 3
Batch: 2022-2026	**Program/Dept.:** ECE	**Section:** ECE-A, B & C
Regulation: R21	**Faculty: Ms.B.Sandhya**	

Question Bank

TLO. No:	Unit No:	Question No	QUESTION BANK	CO	BL	Marks per Q
Module 1:Digital Image Processing						
1.	I	1.	(a) What is meant by digital image processing? Explain how digital images can be represented.	1	1	06
			(b) What are the components of an image processing system.	1	1	06
2	I	2	(a) Explain the fundamental steps involved in typical digital image processing.	1	2	10
			(b) List the applications of digital image processing.	1	2	02
3	I	3	(a) Describe image acquisition and the key factors that affect it.	1	2	07
			(b) List the types of sensors in image acquisition.	1	2	05
4	I	4	(a) Discuss image sampling and quantization with suitable illustrations.	1	3	06
			(b) Explain about the basic relationships and distance measures between pixels in a digital image with example.	1	3	06
5	I	5	(a) Image geometry transformations such as scaling, translation, and rotation.	1	3	06
			(b) Write a detailed note on gray level transformations and their applications.	1	3	06
6	I	6	(a) Derive and explain 2D DFT and its properties.	1	4	06
			(b) Compute the 2D DFT of the given image segment: $[A]=\begin{pmatrix}100 & 0\\ 15 & 255\end{pmatrix}$ Illustrate how the Fourier transform is used for image enhancement by applying a suitable frequency domain filter sequence. Obtain DFT transformed matrix for the given 2x2 image matrix.	1	4	06

<table>
<tr><th>TLO . No:</th><th>Unit No:</th><th>Question No</th><th>QUESTION BANK</th><th>CO</th><th>BL</th><th>Marks per Q</th></tr>
<tr><td>7</td><td>I</td><td>7</td><td>(a) Explain Hadamard and Walsh transforms in terms of image analysis and their applications.
(b) F(x) = {1,2,0,3}. Apply Walsh & Hadamard transform.</td><td>1</td><td>4</td><td>6+6</td></tr>
<tr><td rowspan="2">8</td><td rowspan="2">I</td><td rowspan="2">8</td><td>(a)Explain the Discrete Cosine Transform (DCT) and its use in image compression.</td><td>1</td><td>5</td><td>06</td></tr>
<tr><td>(b)Prove discrete cosine transform is orthogonal for N=4.</td><td>1</td><td>5</td><td>06</td></tr>
<tr><td>9</td><td>I</td><td>9</td><td>Discuss the differences among various image transforms used in image processing.</td><td>1</td><td>4</td><td>12</td></tr>
<tr><td>10</td><td>I</td><td>10</td><td>Highlight the practical applications of image transforms in real-time image analysis.</td><td>1</td><td>3</td><td>12</td></tr>
<tr><td colspan="7">Module - 2: Image Enhancement</td></tr>
<tr><td>13.</td><td>II</td><td>1.</td><td>Discuss arithmetic and logical operations in image enhancement with examples.</td><td>2</td><td>3</td><td>12</td></tr>
<tr><td>14.</td><td>II</td><td>2.</td><td>Explain the concept of point processing and its types with examples.</td><td>2</td><td>2</td><td>12</td></tr>
<tr><td>15.</td><td>II</td><td>3</td><td>Explain the following intensity transformations:

(a) Grey Level Slicing

(b) Bit Plane Slicing</td><td>2</td><td>2</td><td>6+6</td></tr>
<tr><td>16.</td><td>II</td><td>4</td><td>(a) Explain the term Histogram specification or histogram matching.

(b) Explain spatial domain method for enhancement of images.</td><td>2</td><td>2</td><td>6+6</td></tr>
<tr><td>17.</td><td>II</td><td>5</td><td>(a) Explain the term Histogram processing and Histogram equalisation.

(b) Equalize the given histogram.
<table>
<tr><td>Gray Level</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr>
<tr><td>No. of Pixels</td><td>790</td><td>1023</td><td>850</td><td>656</td><td>329</td><td>243</td><td>122</td><td>81</td></tr>
</table></td><td>2</td><td>2</td><td>6+6</td></tr>
<tr><td>18.</td><td>II</td><td>6</td><td>(a)Perform Histogram equalization of the 4x4 image.
$\begin{pmatrix} 4 & 4 & 4 & 3 \\ 3 & 4 & 5 & 4 \\ 3 & 5 & 5 & 4 \\ 3 & 4 & 5 & 4 \end{pmatrix}$
(b) Explain histogram stretching with example.
<table>
<tr><td>Gray Level</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr>
<tr><td>Number of pixels</td><td>0</td><td>0</td><td>50</td><td>60</td><td>50</td><td>20</td><td>10</td><td>0</td></tr>
</table>
Perform histogram stretching so that the new image has a dynamic range of (0,7)</td><td>2</td><td>3</td><td>6+6</td></tr>
<tr><td>19.</td><td>II</td><td>7.</td><td>Explain smoothing filters used in image enhancement in spatial domain and their effects on image.</td><td>2</td><td>2</td><td>12</td></tr>
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20.	II	8.	Explain sharpening filters in the spatial domain with suitable examples.	2	3	12
21.	II	9.	Discuss frequency domain enhancement techniques in detail.	2	4	12
22.	II	10.	Define and describe the homomorphic filter and its applications.	2	3	12
23.	II	11.	Compare smoothing and sharpening in both spatial and frequency domains.	2	4	12
24.	II	12.	Write a short note on the advantages and limitations of various enhancement techniques.	2	5	12
Module 3: Image Restoration						
25.	III	1.	E**xplain the degradation model used in image restoration.**	3	2	12
26.	III	2.	Explain types of noise models.	3	2	12
27.	III	3.	Describe spatial filtering techniques for noise removal.	3	3	12
28.	III	4.	Explain inverse filtering and how it is applied to restore a degraded image.	3	3	12
29.	III	5.	Explain least mean square filtering.	3	3	12
30.	III	6.	Explain the concepts of Wiener filtering for handling noise and degradation.	3	4	12
31.	III	7.	Explain the concept of order-statistic and adaptive filters with application.	3	4	12
32.	III	8.	(a)Write notes on restoration in the presence of spatial non-invariant noise. b) Given below is a 3x3 image. What will the value of the centre pixel change to when this image is passed through: 1 7 5 6 2 3 1 4 2 i) Arithmetic mean filter ii) Geometric mean filter iii) Harmonic mean filter iv) Max filter v) Min filter	3	4	6+6
33.	III	9.	Discuss the techniques for estimating the degradation function.	3	4	12
34.	III	10.	Explain RGB, YUV, and HSI color models and their applications.	3	2	12
35.	III	11.	What are color complements? How do they help in color image processing?	3	3	12
36.	III	12.	Describe methods of image smoothing and sharpening.	3	3	12
Module 4: Image Compression						
37.	IV	1.	Explain the different types of redundancies (coding, interpixel, and psychovisual) in digital images and how they are utilized in compression.	4	2	12
38.	IV	2.	Describe the steps involved in lossless image compression using Variable Length Coding (VLC). Give an example using Huffman coding.	4	3	12
39.	IV	3.	Discuss the working of LZW (Lempel-Ziv-Welch) coding for lossless image compression with a step-by-step example.	4	3	12
40.	IV	4.	What is bit-plane coding? Explain its significance in image compression and illustrate with a suitable binary image.	4	3	12

TLO. No:	Unit No:	Question No	QUESTION BANK	CO	BL	Marks per Q
41.	IV	5.	Describe Predictive Coding with DPCM (Differential Pulse Code Modulation). Derive the encoder and decoder equations with an example.	4	3	12
42.	IV	6.	Compare and contrast lossless and lossy image compression techniques. Explain with examples where each is preferred.	4	4	12
43.	IV	7.	Explain the concept and mathematical background of Transform Coding. How does the DCT help reduce image data size?	4	3	12
44.	IV	8.	Discuss the process of JPEG compression in detail. Include steps such as DCT, quantization, zigzag scanning, and entropy coding.	4	3	12
45.	IV	9.	Describe the structure and workflow of the JPEG 2000 image compression standard. How is it more efficient than the original JPEG?	4	4	12
46.	IV	10.	List and explain the main differences between predictive coding and transform coding. Include use cases where each is effective.	4	4	12
47.	IV	11.	Write short notes on the following (any two): a) Run-Length Encoding with example b) Arithmetic Coding with example c) Image Compression Ratios	4	2	12
48.	IV	12.	What are the major steps involved in lossy transform coding? Compare the performance of JPEG and JPEG 2000 in terms of quality and compression ratio.	4	5	12
Module 5: Image Segmentation						
49.	V	1.	Explain the concept of image segmentation. What are its objectives and applications in digital image processing?	5	2	12
50.	V	2.	Describe point detection using the second derivative (Laplacian) and explain its role in segmentation with examples.	5	3	12
51.	V	3.	Explain edge detection techniques in image segmentation. Compare Sobel, Prewitt, and Canny edge detectors.	5	4	12
52.	V	4.	Describe line detection using masks. How can horizontal, vertical, and diagonal lines be identified using convolution masks?	5	3	12
53.	V	5.	Discuss thresholding for image segmentation. Explain global thresholding with an example.	5	3	12
54.	V	6.	Differentiate between global and adaptive thresholding techniques. Illustrate with diagrams or pixel examples.	5	4	12
55.	V	7.	Describe the concept of region-based segmentation. How does region growing work? Provide an example.	5	4	12
56.	V	8.	What is region splitting and merging? Explain with algorithmic steps and an example.	5	4	12
57.	V	9.	Compare point, line, and edge detection. In what scenarios would each method be preferred in segmentation?	5	4	12
58.	V	10.	Write short notes on: a) Histogram-based thresholding. c) Gradient operators in edge detection.	5	3	12
59.	V	11.	Explain the challenges in image segmentation. How do noise, contrast, and illumination affect the results?	5	4	12

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